Slide Set #13

# Linear Model with Endogenous Regressors GMM and IV estimation

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Notes

## Intro

### Moment Conditions Models (MCM)

Models provided by economic theory often lead to equilibrium/optimality conditions expressed as one or more equations like

$$\mathrm{E}[\boldsymbol{g}(\cdot, \boldsymbol{\beta}) \,|\, \text{some information}] = \mathbf{0},$$

for some function $\boldsymbol{g}(\cdot, \boldsymbol{\beta})$ of observable economic variable and parameters of interest $\boldsymbol{\beta}$

### Generalized Method of Moments (GMM, Hansen, 1982)

a unifying estimation method suited to estimate MCM. Includes both linear and non linear models, IVs, an much more.

### Instrumental variable regression

it's a special case of the GMM with linear $\boldsymbol{g}(\cdot)$ and IVs. Although in the literature they have been developed before the GMM



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## Notations

Response

$y_i$: the usual response/outcome/independent variable

Regressors

$\boldsymbol{z}_i$ is a vector of $L$ regressors of interest. We are interested in estimating partial effects like $\partial \mathrm{E}[y_i \mid \boldsymbol{z}_i]/\partial \boldsymbol{z}_i$. However, some (or all) of the elements of $\boldsymbol{z}_i$ are endogenous

Exogenous variables

$\boldsymbol{x}_i$ is a vector of $K$ exogenous variables. For example $\boldsymbol{x}_i$ may contain IVs. $\boldsymbol{x}_i$ may also contain non-endogenous regressors

**Remark:** traditionally $\boldsymbol{z}$ is used to denote exogenous variables. Hayashi's book does it different, but there is a reason for this



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## Linear model with endogenous regressors

**Assumption (C1): linearity**

Let $\boldsymbol{z}_i \in \mathbb{R}^L$ be a vector of regressors, $\boldsymbol{\delta} \in \mathbb{R}^L$ a vector of $L$ coefficients, $\{\varepsilon_i\}$ is an unobservable error term, and

$$\begin{aligned} y_i =& \delta_1 + \delta_2 z_{i2} + \delta_3 z_{i3} +, \ldots, + \delta_L z_{iL} + \varepsilon_i \qquad \text{for } i = 1, 2, \ldots, n \\ =& \boldsymbol{z}_i' \boldsymbol{\delta} + \varepsilon_i \end{aligned}$$

$z_{i1} = 1$ for all $i = 1, 2, \ldots, n$ if we have a constant term in the model.

**Assumption (C2): ergodic stationarity**

Let $\boldsymbol{x}_i$ be a $K$-dimensional vector of instruments, let $\boldsymbol{w}_i$ be the unique and nonconstant term of $(y_i, \boldsymbol{z}_i', \boldsymbol{x}_i')$. Assume $\{\boldsymbol{w}_i\}$ is jointly stationary and ergodic.



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Denote

$$\boldsymbol{g}_i \;=\; \boldsymbol{x}_i\varepsilon_i \;=\; \boldsymbol{x}_i(y_i - \boldsymbol{z}_i'\boldsymbol{\delta})$$

**Assumption (C3): orthogonality and moment conditions**

Elements of $\boldsymbol{x}_i$ are **predetermined** in the sense that $\mathrm{E}[x_{ik}\varepsilon_i] = 0$ for all $i, k$; or $\mathrm{E}[\boldsymbol{g}_i] = \mathbf{0}$.

(C3) gives a key tool, the moment equation condition

$$\mathrm{E}[\boldsymbol{x}_i(y_i - \boldsymbol{z}_i'\boldsymbol{\delta})] = \mathbf{0}$$

and this is a system of $K$ moment equations with $L$ unknowns in $\boldsymbol{\delta}$:

$$\begin{pmatrix} \text{eq. } 1: & \mathrm{E}[x_{i1}(y_i - \boldsymbol{z}_i'\boldsymbol{\delta})] & = & 0 \\ \text{eq. } 2: & \mathrm{E}[x_{i2}(y_i - \boldsymbol{z}_i'\boldsymbol{\delta})] & = & 0 \\ \vdots & \vdots & \vdots & \\ \text{eq. } K: & \mathrm{E}[x_{iK}(y_i - \boldsymbol{z}_i'\boldsymbol{\delta})] & = & 0 \end{pmatrix}$$



Notes

**Assumption (C4): rank condition**

The $(K \times L)$ matrix $\mathrm{E}[\boldsymbol{x}_i\boldsymbol{z}_i'] = \boldsymbol{\Sigma}_{\boldsymbol{xz}}$ has rank equal to $L$ (full column rank).

**Proposition (identification)**

Under (C1)–(C4), the parameter vector $\boldsymbol{\delta}$ is uniquely identified.

**Proof.** Based on (C3) we know that there exists a $\boldsymbol{\delta}$ such that

$$\mathrm{E}[\boldsymbol{x}_i(y_i - \boldsymbol{z}_i'\boldsymbol{\delta})] = \mathrm{E}[\boldsymbol{x}_i y_i] - \mathrm{E}[\boldsymbol{x}_i\boldsymbol{z}_i']\boldsymbol{\delta} = \mathbf{0}$$

That is

$$\mathrm{E}[\boldsymbol{x}_i\boldsymbol{z}_i']\boldsymbol{\delta} = \mathrm{E}[\boldsymbol{x}_i y_i] \tag{13.1}$$

Based on (C2) we know that these moments exists and do not depend on $i$. From linear algebra we know that the there exists a unique $\boldsymbol{\delta}$ satisfying (13.1) if and only if $\mathrm{rank}(\mathrm{E}[\boldsymbol{x}_i\boldsymbol{z}_i']) = L$. Based on (C4) the unknown parameter $\boldsymbol{\delta}$ is uniquely determined based on moments obtained from the joint distribution of the observables. Therefore, $\boldsymbol{\delta}$ is identified. ■



Notes

$\text{rank}(\boldsymbol{\Sigma}_{\boldsymbol{xz}}) \leq \min\{K, L\}$. Therefore, identifiability can be achieved if $K \geq L$ (order condition). We distinguish the following 3 cases:

Under–identified model
order condition is not satisfied because $K < L$. It's hopeless to estimate $\boldsymbol{\delta}$ because it's not identified.

Just–identified model
$K = L$, rank condition is satisfied. The model is identified and we can estimate $\boldsymbol{\delta}$

Over-identified model
$K > L$, rank condition is satisfied. The model is identified and we can estimate $\boldsymbol{\delta}$ $\boldsymbol{\delta}$



Notes

## Empirical version of the moment conditions (C3)

Define $\sigma_{\boldsymbol{xy}} = \mathrm{E}[\boldsymbol{x}_i y_i]$, and recall that $\boldsymbol{\Sigma}_{\boldsymbol{xz}} = \mathrm{E}[\boldsymbol{x}_i \boldsymbol{z}_i']$. Take their sample mean counterpart

$$\boldsymbol{s}_{\boldsymbol{xy}} = \frac{1}{n}\sum_{i=1}^{n} \boldsymbol{x}_i y_i \qquad \text{and} \qquad \boldsymbol{S}_{\boldsymbol{xz}} = \frac{1}{n}\sum_{i=1}^{n} \boldsymbol{x}_i \boldsymbol{z}_i'$$

Rewrite the sample version of $\mathrm{E}[\boldsymbol{g}_i] = 0$

$$\begin{aligned}
\boldsymbol{g}_n(\boldsymbol{\delta}) =& \frac{1}{n}\sum_{i=1}^{n} \boldsymbol{x}_i (y_i - \boldsymbol{z}_i'\boldsymbol{\delta}) \\
=& \frac{1}{n}\sum_{i=1}^{n} \boldsymbol{x}_i y_i - \left(\frac{1}{n}\sum_{i=1}^{n} \boldsymbol{x}_i \boldsymbol{z}_i'\right)\boldsymbol{\delta} \\
=& \boldsymbol{s}_{\boldsymbol{xy}} - \boldsymbol{S}_{\boldsymbol{xz}}\boldsymbol{\delta}
\end{aligned}$$



Notes

Replace $\mathrm{E}[\cdot]$ with $\frac{1}{n}\sum_{i=1}^{n}$) we obtain the empirical version of the system of moment equations

$$\boldsymbol{g}_n(\boldsymbol{\delta}) = \frac{1}{n}\sum_{i=1}^{n} \boldsymbol{x}_i(y_i - \boldsymbol{z}_i'\boldsymbol{\delta}) = \mathbf{0}$$

In fact this is a vector of $K$ equations of the form

$$\begin{cases} \text{eq. } 1: & \frac{1}{n}\sum_{i=1}^{n} x_{i1}(y_i - \boldsymbol{z}_i'\boldsymbol{\delta}) & = & 0 \\ \text{eq. } 2: & \frac{1}{n}\sum_{i=1}^{n} x_{i2}(y_i - \boldsymbol{z}_i'\boldsymbol{\delta}) & = & 0 \\ & \vdots & & \vdots \\ \text{eq. } K: & \frac{1}{n}\sum_{i=1}^{n} x_{iK}(y_i - \boldsymbol{z}_i'\boldsymbol{\delta}) & = & 0 \end{cases}$$



Notes

## GMM estimation

**Definition (GMM estimator for just–identified linear models)**

Assume $K = L$, the GMM estimator is the vector

$$\hat{\boldsymbol{\delta}} \quad \text{such that} \quad \boldsymbol{g}_n(\hat{\boldsymbol{\delta}}) = \mathbf{0} \tag{13.2}$$

that is, $\hat{\boldsymbol{\delta}}$ solves the empirical version of the moment conditions.

**Remarks**

- The GMM is "*natural estimator*" in all those situations when a model can be specified in terms of a set moment equations
- The GMM framework is easily extended to nonlinear regression models
- When $K > L$ there is a more general GMM definition (which includes (13.2))



Notes

**Proposition (existence and uniqueness of the GMM estimator)**

Assume (C2) and (C4), suppose $K = L$ (just-identified). For $n \to +\infty$, with probability approaching 1, there exists a unique GMM estimator

$$\hat{\boldsymbol{\delta}} \;=\; \boldsymbol{S}_{\boldsymbol{xz}}^{-1}\boldsymbol{s}_{\boldsymbol{xy}} \tag{13.3}$$

**Proof.** Since, $K = L$ by (C4) $\boldsymbol{\Sigma}_{\boldsymbol{xz}}$ is square invertible. By (C2) $\boldsymbol{S}_{\boldsymbol{xz}} \xrightarrow{\text{as}} \boldsymbol{\Sigma}_{\boldsymbol{xz}}$, meaning that $\boldsymbol{S}_{\boldsymbol{xz}}$ is invertible with probability $1$ asymptotically with $n$.

The GMM must satisfy the definition (13.2)

$$\boldsymbol{g}_n(\hat{\boldsymbol{\delta}}) = \boldsymbol{s}_{\boldsymbol{xy}} - \boldsymbol{S}_{\boldsymbol{xz}}\hat{\boldsymbol{\delta}} = \boldsymbol{0} \quad \Longrightarrow \quad \boldsymbol{S}_{\boldsymbol{xz}}\hat{\boldsymbol{\delta}} = \boldsymbol{s}_{\boldsymbol{xy}}$$

When $n \to +\infty$ $\boldsymbol{S}_{\boldsymbol{xz}}^{-1}$ exists with probability 1. Hence,

$$\boldsymbol{S}_{\boldsymbol{xz}}\hat{\boldsymbol{\delta}} = \boldsymbol{s}_{\boldsymbol{xy}} \quad \Longrightarrow \quad \boldsymbol{S}_{\boldsymbol{xz}}^{-1}\boldsymbol{S}_{\boldsymbol{xz}}\hat{\boldsymbol{\delta}} = \boldsymbol{S}_{\boldsymbol{xz}}^{-1}\boldsymbol{s}_{\boldsymbol{xy}} \quad \Longrightarrow \quad \hat{\boldsymbol{\delta}} = \boldsymbol{S}_{\boldsymbol{xz}}^{-1}\boldsymbol{s}_{\boldsymbol{xy}}$$

The proof is completed by noting the inverse is unique. ■



Notes

## Matrix formulation of $\hat{\boldsymbol{\delta}}$

$$\hat{\boldsymbol{\delta}} \;=\; \boldsymbol{S}_{\boldsymbol{xz}}^{-1}\boldsymbol{s}_{\boldsymbol{xy}} \;=\; \left(\frac{1}{n}\sum_{i=1}^{n}\boldsymbol{x}_i\boldsymbol{z}_i'\right)^{-1}\left(\frac{1}{n}\sum_{i=1}^{n}\boldsymbol{x}_i y_i\right)$$

$\sum_{i=1}^{n}$ can be translated into convenient matrix computations (as for OLS)

Let $\boldsymbol{Z}$ be the data matrix about the $L$ regressors, and $\boldsymbol{X}$ be the data matrix of the exogenous variables

$$\boldsymbol{Z} = \begin{pmatrix} z_{11} & z_{12} & \dots & z_{1L} \\ z_{21} & z_{22} & \dots & z_{2L} \\ \vdots & \vdots & \ddots & \vdots \\ z_{n1} & z_{n2} & \dots & z_{nL} \end{pmatrix} \qquad \boldsymbol{X} = \begin{pmatrix} x_{11} & x_{12} & \dots & x_{1,K} \\ x_{21} & x_{22} & \dots & x_{2,K} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \dots & x_{n,K} \end{pmatrix}$$

It can be easily verified (do homeworks) that

$$\hat{\boldsymbol{\delta}} = (\boldsymbol{X}'\boldsymbol{Z})^{-1}\boldsymbol{X}'\boldsymbol{y}$$



Notes

## Obtain a just identified model ($K = L$) using IVs

A common situation:

- some of the regressors are endogenous
- we have found an IV for each endogenous variable

Suppose we have $L$ regressors including the intercept. Assume that $q$ of them are endogenous, while the remaining are exogenous

$$\boldsymbol{z}_i = (1, z_{i,2}, z_{i,3}, \ldots, z_{i,(L-q)},\ z_{i,(L-q+1)}, \ldots, z_{iL})'$$

Suppose we found an IV for each endogenous regressor, suppose they are $\{x_{i1}, x_{i2}, \ldots, x_{iq}\}$. Then we can build the following vector of exogenous variables

$$\boldsymbol{x}_i = (1, z_{i,2}, z_{i,3}, \ldots, z_{i,(L-q)},\ x_{i1}, x_{i2}, \ldots, x_{iq})'$$



Notes

An exogenous regressor can be instrumented by itself
the vector of $K$ exogenous variables is obtained as follows: we retain the $L - q$ exogenous regressor we already had, and we replace the endogenous regressors with our $q$ IVs

*Valid IVs* are *exogenous* $\implies$ we achieve (C3)
recall that a requirement for *validity* of IV is *exogeneity*, i.e. uncorrelation with the regression error $\varepsilon$. It can be easily seen that now our $\boldsymbol{x}_i$ fulfills (C3)

*Valid IVs* are *relevant* $\implies$ we achieve (C4)
*valid* IVs must also be *relevant*, i.e. instruments directly correlate with endogenous regressors explaining a large portion of their variance. It's not straightforward, but when this happens (C4) is fulfilled



Notes

### IV estimator ($\hat{\boldsymbol{\delta}}_{IV}$)

When the vector $\boldsymbol{x}_i$ of exogenous variables is built in this way using valid IVs replacing endogenous regressors, the GMM estimator $\hat{\boldsymbol{\delta}}$ coincides with the well known "*IV estimator*" $\hat{\boldsymbol{\delta}}_{IV}$.

### An old important contribution

The idea of the IV estimator dates back to Wright (1928), and much later became a pillar in causal studies also outside econometrics

### GMM vs IV estimator

they are the same estimator when

- $\boldsymbol{g}(\cdot)$ is linear in the parameter $\boldsymbol{\delta}$ (linear model like C1)
- $K = L$, i.e. the case of just-identified models
- exogenous variables $\boldsymbol{x}$ are obtained replacing with endogenous regressors with IV



Notes

## Statistical properties of the GMM estimator

**Assumption (C5): martingale dependence**

$\{\boldsymbol{g}_i\}$ is a MDS. The $(K \times K)$ matrix of cross moments, $\boldsymbol{S} = \mathrm{E}[\boldsymbol{g}_i \boldsymbol{g}_i']$ is non singular.

### Consistency

under (C1)-(C4), $\hat{\boldsymbol{\delta}} \xrightarrow{\mathrm{p}} \boldsymbol{\delta}$

### Asymptotic Normality

under (C1)–(C5), $\sqrt{n}(\hat{\boldsymbol{\delta}} - \boldsymbol{\delta}) \xrightarrow{\mathrm{d}} \mathcal{N}(\boldsymbol{0}, \mathrm{AVar}(\hat{\boldsymbol{\delta}}))$

### Asymptotic Variance

under (C1)–(C5), $\mathrm{AVar}(\hat{\boldsymbol{\delta}}) = \boldsymbol{\Sigma}_{\boldsymbol{xz}}^{-1} \boldsymbol{S} \boldsymbol{\Sigma}_{\boldsymbol{xz}}^{-1}$

**Note:** for the previous properties see the homeworks file



Notes

## Estimation of the asymptotic variance

Residuals

$$e_i = y_i - \boldsymbol{z}_i'\hat{\boldsymbol{\delta}}$$

**Assumption (C6): 2nd moments of $\boldsymbol{z}$**

We assume that $\mathrm{E}[|\boldsymbol{z}_i\boldsymbol{z}_i'|] < \infty$

Consistent estimator for the unconditional error variance
under (C1), (C2), and (C6), it can be shown that

$$s^2 \;=\; \frac{1}{n}\sum_{i=1}^{n} e_i^2 \xrightarrow{\mathsf{p}} \mathrm{E}[\varepsilon_i^2] \;=\; \mathrm{Var}[\varepsilon_i]$$



Notes

**Assumption (C7): existence of higher order cross-moments**

$\mathrm{E}[(x_{ik}z_{il})^2] < \infty$, for all $k = 1, 2, \ldots, K$ and $l = 1, 2, \ldots, L$.

(C7) implies that the joint distribution of regressors and instruments has a "nice" tail behavior

Consistent estimation of $\boldsymbol{S}$
under (C1), (C2) and (C7):

$$\hat{\boldsymbol{S}} = \frac{1}{n}\sum_{i=1}^{n} e_i^2 \boldsymbol{x}_i\boldsymbol{x}_i' \xrightarrow{\mathsf{p}} \boldsymbol{S}$$

Consistent estimator for $\mathrm{AVar}(\hat{\boldsymbol{\delta}})$
under (C1), (C2) and (C7):

$$\widehat{\mathrm{AVar}\,(\boldsymbol{\delta})} = \boldsymbol{S}_{\boldsymbol{xz}}^{-1}\hat{\boldsymbol{S}}\boldsymbol{S}_{\boldsymbol{xz}}^{-1} \xrightarrow{\mathsf{p}} \mathrm{AVar}(\boldsymbol{\delta})$$

**Note:** for the previous properties see the homeworks file



Notes

### Approximate asymptotic variance of $\hat{\boldsymbol{\delta}}$

for sufficiently large $n$ the variance of $\hat{\boldsymbol{\delta}}$ can be approximated as

$$\widehat{\mathrm{Var}[\boldsymbol{\delta}]} \overset{\mathsf{a}}{\approx} (\boldsymbol{X}'\boldsymbol{Z})^{-1}\boldsymbol{X}'\boldsymbol{B}\boldsymbol{X}(\boldsymbol{X}'\boldsymbol{Z})^{-1}$$

(see homeworks file)

### Inference using $\hat{\boldsymbol{\delta}}$

- standard errors: same as for (B)-model, but using $\widehat{\mathrm{AVar}\,(\boldsymbol{\delta})}$
- tests on individual coefficients: same as for (B)-model, but using $\widehat{\mathrm{AVar}\,(\boldsymbol{\delta})}$
- Wald's tests for linear and nonlinear restrictions: same as for (B)-model, but using $\widehat{\mathrm{AVar}\,(\boldsymbol{\delta})}$



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## Concluding remarks

### Recap

When $K = L$, and $\boldsymbol{x}$ contains valid instruments, the GMM for model (C1)–(C3) is equivalent to the IV estimator

### Conditional heteroscedasticity

As for the (B)-model framework, the (C)-model framework can accommodate conditional heteroscedasticity. Therefore, all inference based on the asymptotic distribution of $\hat{\boldsymbol{\delta}}$ is resistant to conditional heteroscedasticity

### IV, 2SLS and conditional homoscedasticity

*if we assume homoschedastic errors* this GMM/IV estimator becomes equivalent to "*2SLS (2-stages least squares)*". 2SLS is another classical and popular estimator based on the idea of using IV variables.



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